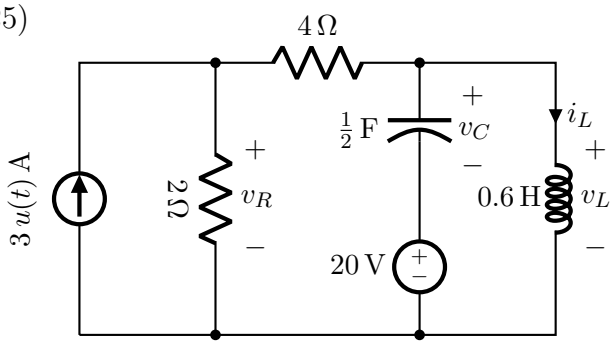
EE2015: Electric Circuits and Networks

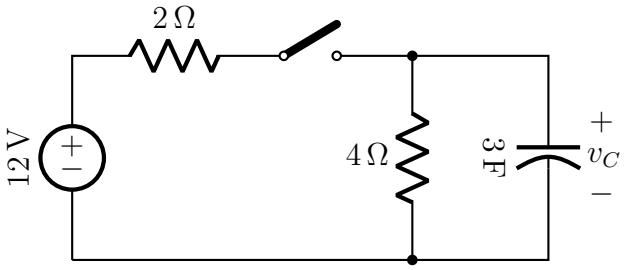
Tutorial 6

(September 26, 2025)

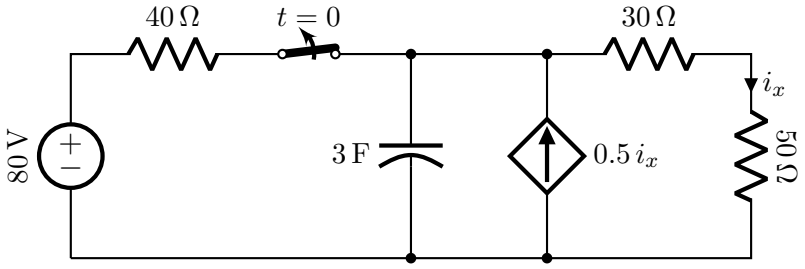
1. For the circuit shown on the right find: (a) $i_L(0^+)$, $v_C(0^+)$, $v_R(0^+)$, (b) $i'_L(0^+)$, $v'_C(0^+)$, $v'_R(0^+)$, and (c) $i_L(\infty)$, $v_C(\infty)$, $v_R(\infty)$.



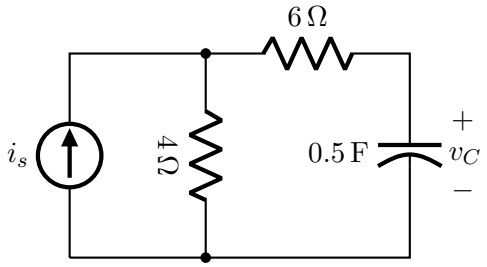
2. For the circuit shown on the left find $v_C(t)$ if
   (a) The switch has been open for a long time and is closed at $t = 0$.
   (b) The switch has been closed for a long time and is opened at $t = 0$.



3. For the circuit shown on the right find $i_x(t)$ for all t.



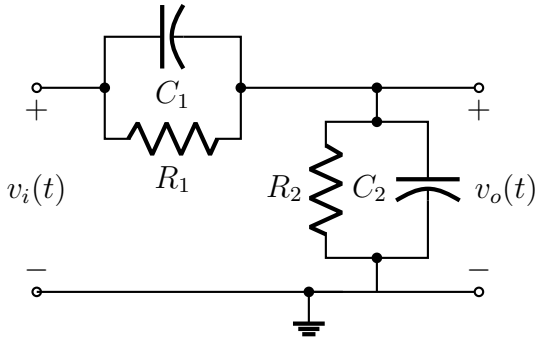
4. For the circuit shown on the left it is given that $v_C(0^-) = 0$. A current pulse $i_s(t) = 2 \cdot [u(t) - u(t-1)]$ A is applied to the circuit. Find $v_C(t)$ for $t \geq 0$.

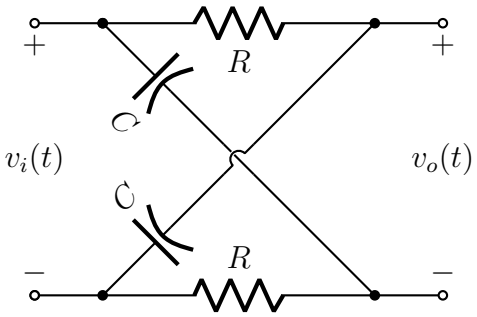


5. (a) A voltage source $v_s(t) = e^{-2t}\,u(t)$ is applied across a series combination of $1\,\Omega$ and $1\,F$. Find $v_c(t)$ for $t \geq 0$.
   (b) Repeat for $v_s(t) = e^{-t}\,u(t)$. Compare this answer with the previous one and comment on the result.

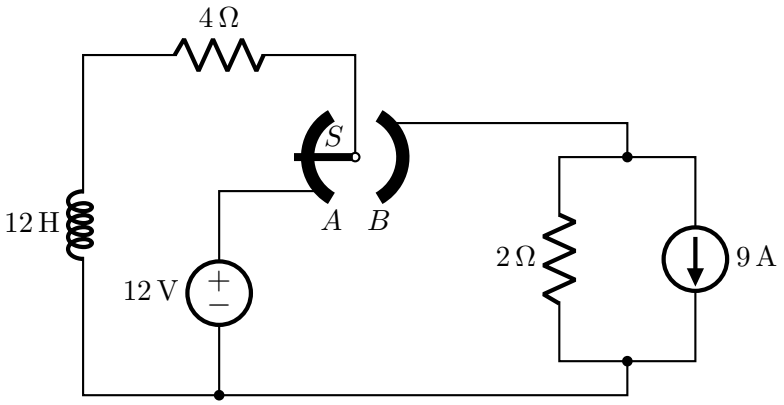
6. Consider the two circuits shown below. Derive the expressions for $v_o(t)$ when $v_i(t) = V_p\,u(t)$. The capacitors have zero initial charge. Plot $v_o(t)$ roughly to scale and mark

the important points, including the time-constants. Next, open-circuit the capacitors and calculate $v_o(t)$. How are these values related to $v_o(t)$ calculated previously?



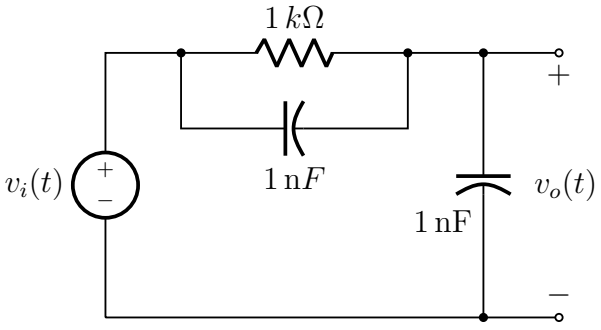


7. In the circuit shown below, the rotary switch S rotates at 10 rpm. It makes contact alternately with the conducting segments A and B for 3 s each. The transition from A to B happens intantaneously. Assume that S has been rotating for a long time.
   (a) Plot $i_L(t)$, clearly marking the maximum and minimum values.
   (b) Calculate the maximum and minimum voltage across the 2 Ω resistance, and the total energy consumed by this resistance during each revolution of S.



8. The circuit shown on the right is excited by $v_i(t)$, where $v_i(t) = \sum_{k=0}^{\infty} x(t-kT_0)$. The waveform $x(t)$ is defined as follows:

$$x(t) = \begin{cases} 1 & 0 < t < T_0/2 \\ -1 & T_0/2 < t < T_0 \\ 0 & \text{otherwise} \end{cases}$$



where $T_0 = 2\,\mu$s. Sketch $v_o(t)$ after steady-state has been reached and mark the important points.